

$$\Delta y_{SEj} = \sum_i^{AB(n=11)} y_{0ij} * [\exp(\beta_{SEj}) - 1] * Population_i * SmokeEventDays_i$$

where i = air basin, j = outcome group, β_{SE} = overall effect estimate for a smoke event day, y_0 = baseline incidence rate of EDVs, SmokeEventDays = number of smoke event days during the study period, and Δy_{SE} = total number of EDVs attributable to smoke events. It should be noted that β_{SE} is the effect estimate for the binary smoke event indicator.

Outcome data were cleaned and prepared using SAS version 9.4. All other analyses were conducted in RStudio using R version 4.2.1 (R Core Team, 2021; RStudio Team, 2020).

3. Results

3.1. Descriptive statistics of exposure data

Across the 15 air basins in California, daily wildfire-specific PM_{2.5} concentrations from June–December 2016–2019 ranged from 0 to 164.6 $\mu\text{g}/\text{m}^3$ (median = 0 $\mu\text{g}/\text{m}^3$, mean = 1.1 $\mu\text{g}/\text{m}^3$), with 95% of air basin-days having concentrations below 5 $\mu\text{g}/\text{m}^3$. There were 257 smoke events out of 12,840 air basin-days. Four of the air basins (South Coast, San Diego, Salton Sea, and Mojave Desert) were excluded from this study, since they did not have any smoke events during the study period, based on our definition (Fig. 1).

Among the remaining 11 air basins, the median wildfire-specific PM_{2.5} concentration for smoke events was 23.1 $\mu\text{g}/\text{m}^3$ (interquartile range (IQR): 14.9), compared to a median of 0 $\mu\text{g}/\text{m}^3$ (IQR: 0.2) on days that were not smoke events (non-smoke events). The mean number of smoke events per basin was 23.4. The air basins with the fewest and greatest number of smoke events were North Central Coast ($n = 7$) and the Northeast Plateau ($n = 61$), respectively (Fig. 1).

3.2. Description of study population and outcome data

The 11 air basins with smoke events during the study period included 17,847,917 people, which is about 45.4% of California's population. The study population had a similar age group and sex distribution to that of the entire state (Table 2). With respect to race/ethnicity composition, the study population had a slightly higher percentage of people who were White (42.6% vs 37.2%) or API (16.2% vs 14.6%) and a lower percentage of people who were Hispanic (32.1% vs 39.0%), compared to the breakdown for the entire state.

The number of EDVs totaled 1,363,054 for all respiratory diseases, 1,166,893 for all cardiovascular diseases, 146,888 for diabetes, and 630,113 for all mental health outcomes (Table 3). The most common disease subgroups for respiratory diseases, cardiovascular diseases, and mental health disorders were AURI (45.7% of all respiratory EDVs), dysrhythmia (10.7% of all cardiovascular EDVs), and substance use disorders (36.1% of mental health EDVs), respectively. The mean daily rate of EDVs per 100,000 people was higher for smoke events compared to non-smoke events for most respiratory disease categories in this study. For all other outcomes, the rate of EDVs was similar between smoke events vs non-smoke events.

3.3. Associations between smoke events and EDVs

Smoke events were associated with EDVs for several disease outcomes, with the greatest associations observed for respiratory outcomes (Fig. 2, Fig. S1, Table S2). The lag that resulted in the best model fit differed by disease subgroup, but results were generally consistent across lags for most outcomes. Smoke events were associated with a 14.3% (95% CI: 6.8, 22.5%) increase in the risk of EDVs for all respiratory diseases at lag 1. Smoke events showed positive associations with all respiratory disease subgroups, with the strongest increase in the risk for asthma EDVs at lag 0 (57.1%, 95% CI: 44.5, 70.8%). EDVs for

pneumonia and AURI showed positive associations with smoke events, although they were not statistically significant (Table S2).

Smoke events were also associated with a 3.2% (95% CI: 1.7, 4.7%) increase in the risk of EDVs for all cardiovascular diseases at lag 10. The association between smoke events and EDVs for all cardiovascular diseases was negative or null during earlier lags but gradually became positive with longer lags, suggesting that cardiovascular impacts may be delayed compared to respiratory impacts. No significant associations for specific cardiovascular disease subgroups were observed (Fig. S1). The association between smoke events and diabetes EDVs was positive, but not statistically significant. With regards to mental health EDVs, smoke events were positively associated with EDVs for schizophrenia at lag 0 (13.1% increase in risk, 95% CI: 5.1, 21.8%) but significantly negatively associated with EDVs for substance use disorders and suicide (Fig. 2, Fig. S1, Table S2). The associations for schizophrenia and suicide were consistent across short-term lags through day 2, but changed directions with longer lags tested. The negative association for substance use disorders was observed in other lags tested, but the association was only statistically significant for lag 10, suggesting that this result may be more sensitive to the specific lag (Fig. 2).

Our results were generally robust to adjustment for co-pollutants (Fig. S2–S5). The models with and without non-wildfire PM_{2.5} included 11 air basins, the same dataset as in the main model. Analyses for O₃, NO₂, and CO included fewer air basins (eight, five, and six, respectively) due to insufficient monitor coverage across the state. Adjustment for CO or non-wildfire PM_{2.5} resulted in greater reductions in the magnitude of associations, particularly for respiratory outcomes, but the results were still similar to unadjusted estimates (Fig. S2, S5). Given the fewer number of air basins included in some co-pollutant analyses, the results may not be as representative of the entire state.

We also assessed the robustness of our results to different smoke event definitions by comparing results based on the overall threshold to those using basin-specific thresholds. For the same subset of 11 air basins, we found similar results between these two threshold definitions (Fig. S6). For some outcomes, namely pneumonia, mood disorders, and schizophrenia, results based on basin-specific thresholds led to a slightly higher increase in the risk of EDVs. We also observed similar results based on basin-specific thresholds with and without the four basins excluded from the main analysis, although the increase in the risk of asthma EDVs was lower when all basins were included (Fig. S6). This observation was likely driven by the four basins that were excluded in the main analysis, since the first-stage model estimates for asthma EDVs for these four basins were close to null (Fig. S7).

Using a health impact function, we calculated the health burden attributable to smoke events, based on the relative risk estimates we

Table 2
Demographic characteristics of the entire state and the study population.

Demographic group	Entire state	Study population
N	39,275,325	17,847,917
Age group (years) (%)		
0–17	22.97	23.04
18–64	63.06	62.31
65+	13.97	14.66
Race/ethnicity (%)		
White	37.18	42.62
Black	5.52	4.71
Hispanic	39.02	32.12
API	14.64	16.25
Other	3.64	4.30
Sex (%)		
Male	49.70	49.81
Female	50.30	50.19

The Other subgroup was not included in our analysis stratified by race/ethnicity.

Table 3

Mean daily rate of EDVs per 100,000 people for each outcome on all days, smoke event days, and non-smoke event days.

Outcome Group	N	Mean daily rate of EDVs per 100,000 people		
		All days	Smoke event days	Non-smoke event days
All respiratory diseases	1,363,054	8.92	10.03	8.90
Asthma	163,994	1.07	1.77	1.06
AURI	622,512	4.07	4.40	4.07
CLRD	102,189	0.67	0.74	0.67
Pneumonia	143,820	0.94	0.92	0.94
All cardiovascular diseases	1,166,893	7.64	7.55	7.64
AMI	59,686	0.39	0.41	0.39
Cerebrovascular disease	83,924	0.55	0.54	0.55
Dysrhythmia	124,935	0.82	0.82	0.82
IHD	86,699	0.57	0.59	0.57
PVD	67,111	0.44	0.44	0.44
TIA	27,261	0.18	0.17	0.18
Diabetes	146,888	0.96	0.99	0.96
All mental health disorders	630,113	4.12	4.03	4.13
Anxiety	162,732	1.07	1.03	1.07
Mood disorders	72,896	0.48	0.46	0.48
Schizophrenia	66,577	0.44	0.46	0.44
Substance use disorders	227,765	1.49	1.44	1.49
Suicide	83,584	0.55	0.52	0.55

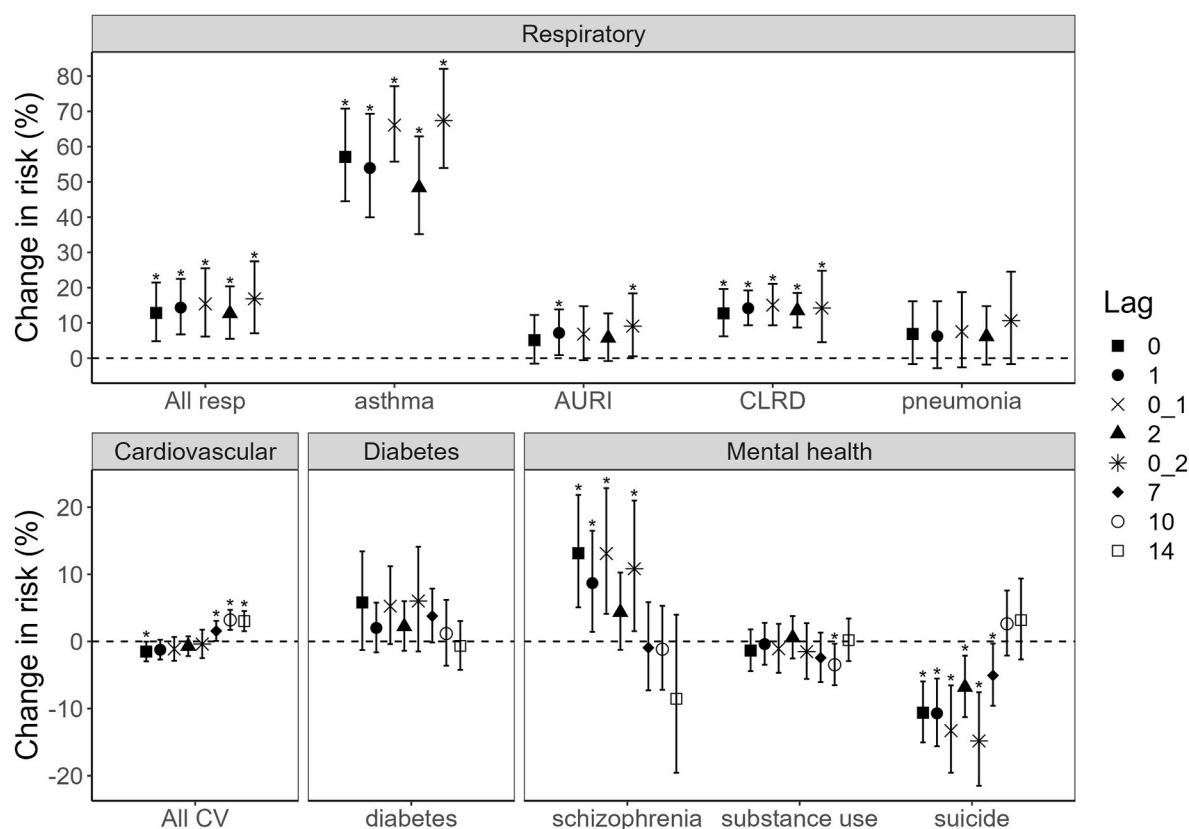


Fig. 2. Percent change in risk of EDVs for select respiratory, cardiovascular, diabetes, and mental health diagnoses associated with smoke events.

The lag with the best model fit is as follows: All resp – lag 1, asthma – lag 0, AURI – lag 2, CLRD – lag 0, pneumonia – lag 0_2, All CV – lag 10, diabetes – lag 0_2, schizophrenia – lag 0, substance use – lag 10, and suicide – lag 0_1. All resp: all respiratory diseases; All CV: all cardiovascular diseases. Bars: 95% confidence intervals. Dotted line represents no change in risk. *p < 0.05.

obtained (Table S2). The estimated health burden attributable to smoke events was greatest for all respiratory diseases combined, with 4597 (95% CI: 2164, 7203) attributable EDVs. There were 2204 (95% CI: 1718, 2733) attributable EDVs for asthma and 323 (95% CI: 158, 498)

attributable EDVs for CLRD. We estimated 889 (95% CI: 476, 1307) attributable EDVs for all cardiovascular diseases combined. There were an estimated 208 (95% CI: 81, 345) attributable EDVs for schizophrenia.

3.4. Subpopulations at increased risk of EDVs associated with smoke events

Stratification analyses by race/ethnicity identified a few statistically significant differences among subpopulations (Fig. 3, Table S3). The risk of diabetes EDVs at lag 0.2 for people who are Hispanic (13.7%, 95% CI: 2.2, 26.4%) was significantly higher than the risk for people who are White (−2.6%, 95% CI: −9.8, 5.2%) (Fig. 3A, Table S3). The risk of diabetes EDVs was trending positive among people who are Black or API, but we did not observe a significant difference compared to the risk for people who are White. For anxiety, we observed a significantly lower risk of EDVs among people who are API compared to people who are White, although neither subgroup had a significant change in risk compared to the null (Fig. 3B, Table S3).

There were also some potential differences that may be worth further investigation in future studies. While there were no significant differences in the risk of asthma EDVs across race/ethnicity subgroups, the risk was trending lower among people who are Black or API (Fig. 3C). For pneumonia, we observed the greatest increase in risk of EDVs for people who are API or Hispanic, although the risk was elevated for all race/ethnicity subgroups (Fig. 3D). There were no significant differences in the risk of EDVs for cardiovascular disease outcomes among race/ethnicity subpopulations (Table S3). However, the risks of EDVs for AMI and dysrhythmia were trending positive for people who are Black (Fig. 3E and F, Table S3).

There were some differential effects by age. People ages 65 and older had a significantly higher risk of EDVs for transient ischemic attacks at lag 2 compared to people ages 18–64 years old, although the confidence interval contained the null value (8.3%, 95% CI: −7.9, 27.4%) (Table S3). Children ages 0–17 years had a significantly lower risk of EDVs for all mental health outcomes at lag 7 (−12.7%, 95% CI: −18.6, −6.4%), compared to the risk for people ages 18–64 years old.

There were no statistically significant differences by sex, but the increases in risk for asthma and CLRD were trending higher among females.

4. Discussion

In this study, we investigated short-term associations between smoke events and EDVs for various causes in California from June–December 2016–2019. Our results are consistent with previous epidemiologic studies in California that found positive associations between wildfire smoke and short-term adverse respiratory effects, most notably for

asthma and CLRD (Aguilera et al., 2021a; Delfino et al., 2009; Heaney et al., 2022; Reid et al., 2016b). While it is difficult to quantitatively compare estimates across studies, given study differences in smoke event thresholds, time periods, and geography, the magnitude of associations we observed for respiratory EDVs (~14% increase) is reasonable, compared to those reported in previous studies. For example, a study during 2011–2017 in San Diego, California found a 30% increase in respiratory EDVs for a 10-unit increase in wildfire-specific PM_{2.5} (Aguilera et al., 2021a), whereas two other studies found a ~3% increase in respiratory EDVs during the 2008 northern California wildfires and for smoke event days during the 2004–2009 wildfire seasons (Heaney et al., 2022; Reid et al., 2016b). We also observed positive associations between smoke events and EDVs for all cardiovascular diseases combined at a later lag but did not find any associations for specific cardiovascular disease subgroups, which is largely consistent with the findings of a recent review (Chen et al., 2021). Our finding of a positive association between smoke events and schizophrenia EDVs also adds to our understanding of mental health effects of wildfire smoke (Eisenman and Galway, 2022; To et al., 2021).

Our observation of strong, immediate respiratory effects and more delayed cardiovascular effects may be partly explained by the timing of the physiological effects of wildfire smoke. Toxicological studies have shown that animals exposed to wildfire smoke initially experience increased inflammation in the lungs, which is followed by inflammation and tissue injury in other organs such as the heart (Black et al., 2017; Kim et al., 2014, 2018). Furthermore, a recent study of mice exposed to wildfire smoke reported changes in the expression of genes involved in hypoxia and cell stress signaling pathways, which have been shown to be involved in tissue injury and inflammation in the heart (Carberry et al., 2022; Giussani et al., 2014). These gene expression changes were first observed in the lungs and were followed by the release of plasma extracellular vesicles (EVs) containing microRNAs (miRNAs) involved in those pathways and changes in expression of the same miRNAs in the heart (Carberry et al., 2022). The earlier effects observed in the lungs are consistent with the strong positive association between smoke events and short-term respiratory impacts that we found. These toxicological findings are also consistent with our finding of a positive association between smoke events and EDVs for cardiovascular diseases at a later lag (lag 10), although we did not detect significant associations with EDVs for any cardiovascular disease subgroups.

Several biological mechanisms of action have been proposed for the cardiorespiratory effects of wildfire smoke. Wildfire smoke inhaled into the lungs is thought to activate neural receptors, leading to changes in

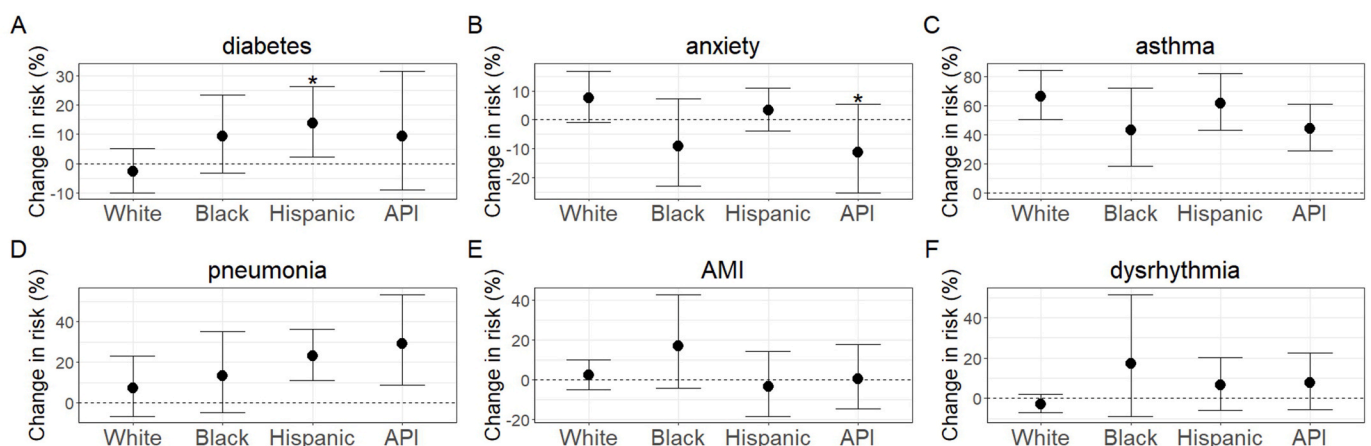


Fig. 3. Percent change in risk of EDVs for (a) diabetes, (b) anxiety, (c) asthma, (d) pneumonia, (e) AMI, and (f) dysrhythmia associated with smoke events, stratified by race/ethnicity.

The lag with the best model fit for each outcome is shown: diabetes – lag 0.2, anxiety – lag 0.1, asthma – lag 0, pneumonia – lag 0.2, AMI – lag 0, and dysrhythmia – lag 1. Bars: 95% confidence intervals. Dotted line represents no change in risk. * $p < 0.05$ for the Z test of interaction.

heart rate and blood pressure (Cascio et al., 2018). Wildfire smoke in the lung has also been shown to increase oxidative stress and systemic inflammation, leading to disruptions in the cell cycle and release of EVs, which play important roles in cell-cell communication. EVs transport molecules including miRNAs, which can alter expression of genes and proteins in target cells (Cascio et al., 2018; Chen et al., 2021). Finally, ultrafine components of wildfire smoke may directly translocate into the bloodstream and reach other organ systems through the circulatory system. Studies have shown that the chemical composition of PM_{2.5} from wildfire smoke has a higher proportion of organic compounds compared to non-wildfire PM_{2.5} (Liu and Peng, 2019; Makkonen et al., 2010; Verma et al., 2009). In addition, smoke-impacted days due to fires in the wildland-urban interface have been observed to have higher concentrations of trace metals and other chemicals harmful to human health, with distinct chemical profiles for particular wildfires (Boaggio et al., 2022; California Air Resources Board, 2021). Future studies on the health effects of wildfire-specific PM_{2.5} constituents may help elucidate the toxicity of wildfire smoke and how this may differ depending on the materials burned.

Associations between smoke events and EDVs for mental health disorders were mixed. We observed positive associations between smoke events and EDVs for schizophrenia and negative associations for EDVs for most of the other mental health outcomes included in this study. Schizophrenia is a mental illness characterized by hallucinations, delusions, and disordered thinking and behavior. One of the disease phenotypes of schizophrenia involves neuroinflammation and alterations to the immune system, which has been observed in mice that were exposed to wildfire smoke-derived particulate matter (Comer et al., 2020; Kroken et al., 2018; Müller and Schwarz, 2010; Scieszka et al., 2022). Other factors that may have played a role include increased psychosocial stress during smoke events, reduced ability to relocate or otherwise reduce smoke exposure, and factors related to medication use, including medication nonadherence that may have exacerbated psychotic symptoms or adverse side effects of antipsychotic medications (Block and Calderón-Garcidueñas, 2009; Bretler et al., 2019; Haddad et al., 2014; McCann et al., 2009). While there have been mixed results from epidemiologic studies on the association between ambient, non-wildfire PM_{2.5} and schizophrenia (Gao et al., 2017; Newbury et al., 2019; Nguyen et al., 2021; Wei et al., 2022), our results indicate that the association between wildfire smoke and schizophrenia is worth further investigation.

For other mental health disorders, our findings were consistent with previous studies using data on hospital visits that found null or negative associations between wildfire smoke and short-term mental health effects (Malig et al., 2021; Moore et al., 2006; Reid et al., 2016a). There are several potential explanations for this finding. The presence of wildfires may have reduced social activities, and consequently, EDVs related to substance use. In addition, patients may have chosen to avoid coming to the hospital for less urgent or non-life-threatening conditions during the initial days following smoke events (Schranz et al., 2010). Hospitals may have also prioritized treating patients for more urgent issues, such as respiratory outcomes. This is consistent with our observation of lower rates of EDVs for most mental health outcomes on days that were smoke events compared to days that were not smoke events (Table 3). Mental health effects also could have exacerbated pre-existing health conditions and contributed to increased visits for respiratory effects or other physical diseases. For example, a survey of a medically vulnerable population in Mariposa County, California found that a respondent's anxiety from the presence of smoke contributed to their cardiovascular symptoms (Hoshiko et al., 2023). It is possible that patients would seek care primarily for acute, physical symptoms rather than potential underlying mental health issues. Qualitative studies may be more suited for understanding the mental health impacts of wildfires, as they extend beyond diagnostic codes and can provide much richer information on how people's mental health and well-being may be affected (Brown et al., 2021; Isaac et al., 2023; Silveira et al., 2021; To

et al., 2021). In addition, long-term exposure may be a more appropriate metric, given the diverse pathways through which wildfire smoke may act, including reduced physical activity, social isolation, uncertainty due to lack of access to timely information, and impacts of evacuation and/or livelihood (Eisenman et al., 2021; Eisenman and Galway, 2022).

Our stratified analyses identified significant effect modification by race/ethnicity or age for some outcomes. We observed a higher risk of EDVs for diabetes among people who are Hispanic and a lower risk of EDVs for anxiety among people who are API. However, we did not observe differences in the risk of EDVs for respiratory outcomes between children and adults, as has been previously reported (Heaney et al., 2022; Kondo et al., 2019). Future studies using more granular age groups, such as children under the age of five, may identify potential differential effects by age (Heaney et al., 2022). Among people who are Black, we observed trending positive associations with EDVs for several cardiovascular disease subgroups, namely AMI and dysrhythmia. People who are Black experience environmental health inequities due to a variety of factors, including but not limited to, higher exposure to air pollution, due to discriminatory redlining practices, racial discrimination in the health care system, and barriers to green space access (Jennings and Johnson Gaither, 2015; Lane et al., 2022; Robinson et al., 2023; Yearby et al., 2022). We also observed trending greater positive associations in the risk of EDVs for pneumonia, AURI, and IHD among people who are API. There may be additional disparities within APIs that were masked by the aggregation of data, since the racial/ethnic category of API consists of great diversity in ethnicities, immigration patterns, socioeconomic status, and health outcomes (Budiman and Ruiz, 2021a, 2021b; Yi et al., 2022). Finally, the increases in the risk of asthma EDVs and of CLRD EDVs were trending higher among females compared to males, which is consistent with what has been reported previously (Kondo et al., 2019; Reid et al., 2016b). Future studies with larger numbers of EDVs and more disaggregated data, including specific subgroups within API, could help better detect potential health disparities associated with smoke events among, and within, subgroups for various outcomes.

This study had several strengths. We used wildfire-specific PM_{2.5} concentrations derived from an ensemble machine learning model, which allowed us to hone in on the specific health effects of smoke events. Using ZCTA-level exposures versus monitor data also provided better spatial coverage across the state. In addition, our study covered multiple fires across recent years in California when wildfire smoke exposure has increased and impacted many populations across the state, including urban areas that were not previously affected by large-scale wildfire smoke exposure. Therefore, our findings are likely to be more generalizable to other areas that experience wildfires with similar wildfire smoke composition. We investigated a wide range of disease categories, including those that were rarely examined in previous wildfire smoke studies, such as schizophrenia. Our study also adds to the few studies that have assessed effect modification by race/ethnicity on the health effects of wildfire smoke.

Our study also had several limitations. We used air basins as our unit of analysis to obtain sufficient case counts to conduct stratified analyses. However, there may be more exposure misclassification with aggregation to a larger geographic area. Since we did not include four air basins in Southern California, our results may be less generalizable to that area. We were unable to account for potential underestimation of case counts due to barriers in accessing healthcare or changes in healthcare-seeking behaviors during an active wildfire. We were also unable to investigate health effects for specific sensitive populations that are disproportionately exposed to wildfire smoke, including undocumented agricultural workers, people who are unhoused, and Indigenous peoples, due to insufficient case numbers and lack of information in our data sources (Méndez et al., 2020). In addition to increased exposure to wildfire smoke, Hispanic or Indigenous farmworkers are disproportionately impacted by racism, inadequate labor protections and job insecurity, linguistic isolation, reduced access to health care, and fear of